

Volatility modelling for Indian markets

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Abstract

This paper forecasts the 30-day realized variance of the NIFTY 50 index using machine learning techniques. Using daily NIFTY options data from 2020-2025, we evaluate linear models and a feed-forward neural network. We also study the variance risk premium, defined as realized variance minus the model-free India VIX, which we compute following [6]. Our results show that ATM and skew-based features of OTM options contains a strong predictive information.

1 Introduction

Forecasting future market volatility is a central task in quantitative finance, with applications in derivative pricing, portfolio risk management, and volatility trading. In equity markets, implied volatilities extracted from option prices provide a forward-looking view of uncertainty, and volatility indices such as the India VIX have become standard benchmarks for expected market variance. Traditionally, these indices are constructed using a variance swap pricing formula under the risk-neutral measure, aggregating a large set of out-of-the-money (OTM) options to infer the expected 30-day variance of the underlying index.

The seminal Black–Scholes (1973)[1] model introduced a constant volatility assumption in a risk-neutral option pricing framework, laying the groundwork for linking option prices to expected market variance via implied volatility. Volatility indices such as the CBOE’s VIX[2] were later introduced to measure the market’s short-term expected variance directly from option prices without relying on a specific model. Although the India VIX has proven informative as a market-implied measure of volatility, it faces some limitations, such as- the index weighting scheme depends on numerous deep OTM options that are often illiquid in the Indian market, making replication costly and noisy. On the other hand, because the VIX-style construction is based on the risk-neutral measure, it may not optimally forecast the realized variance. The difference between these two quantities, called the variance risk premium (VRP), captures compensation investors demand for bearing volatility risk and thus contains valuable information about market sentiment and future volatility dynamics.

Volatility forecasting has been a useful approach for trading for the past many years. There exists many volatility models in the literature. Time-series volatility models such as the ARCH model [3] and its generalization GARCH [4] captured volatility clustering in asset returns, while the HAR model [5] introduced a multi-horizon approach to realized volatility (RV) that accounts for long-memory in the volatility process. Over time, researchers started using forward-looking, option-implied features such as the VIX to improve forecasts of realized variance. Recent studies combine these option-based measures with machine learning techniques, marking a shift toward more flexible, data-driven methods [6]-[12]. For instance, Carr *et al.* [6] use option price data and learning algorithms (e.g. linear regression, neural networks and random forest) to enhance volatility index forecasting by finding a new weighting scheme for the VIX styled index.

This goal of this paper is to develop a volatility forecasting model of the NIFTY index using option data, along the lines of work by Carr, Wu,

and Zhang [6]. In their study, predict the realized variance of the *S&P* 500 index and, in addition, propose a revised weighting scheme for a VIX-style volatility index.

We formulate two supervised regression problems: (1) predicting future realized variance, and (2) predicting the deviation between realized variance and the India VIX, i.e., the variance risk premium. For each task we compare ordinary least squares (OLS), ridge regression, and a feedforward neural network (FNN) with ReLU activation. The models are trained and evaluated using a rolling, out-of-sample (OOS) validation scheme with a 30-day forecast horizon.

The remainder of the paper is organized as follows. Section 2 reviews the construction of the India VIX and outlines its limitations. Section 3 describes the regression framework, feature engineering and residual analysis. In section 4 we describe all the machine learning models employed in this paper. We summarize the results in 5 followed by 6 with concluding remarks and directions for future research.

2 India VIX

India VIX [13] is NSE’s volatility index that measures the market’s expectation of the NIFTY 50’s annualized volatility over the next 30 calendar days. Unlike a price index, India VIX is derived from option order-book quotes (best bid/ask) on NIFTY index options and is expressed in percent per annum.

We use NIFTY50 options data from 2020-07-01 to 2025-05-08 (yyyy-mm-dd) for model building and testing. To define the Variance risk premium (VRP) $VIX^2 - Variance$ we use the historical India VIX data obtained from the National Stock Exchange of India (NSE) which is also verified. The details of the computation are given below.

NSE adapts the CBOE VIX framework to NIFTY options with a few

implementation choices (risk-free rate, forward determination, filtering and interpolation of quotes). For each of two expiries bracketing 30 days (“near” and “next”), NSE forms a strip of out-of-the-money (OTM) puts (for strikes $K_i < K_0$) and OTM calls (for $K_i > K_0$); at-the-money K_0 contributes via the average of call/put midquotes. Let $Q(K_i)$ denote the midpoint of bid and ask for strike K_i , ΔK_i the strike spacing, R the risk-free rate to expiration, and T the time to expiration (in years, computed in minutes). The forward F is taken from the corresponding NIFTY futures; K_0 is the strike just below F .

The (risk-neutral) variance for each tenor is computed as

$$\sigma^2(T) = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2. \quad (1)$$

With near/next maturities (T_1, T_2) producing (σ_1^2, σ_2^2) , the 30-day constant-maturity variance is obtained by linear interpolation on *variance* and then annualized (per a 365-day year) and scaled by 100:

$$\text{India VIX} = 100 \times \sqrt{\frac{T_2 - T}{T_2 - T_1} \sigma_1^2 + \frac{T - T_1}{T_2 - T_1} \sigma_2^2} \quad T = \frac{30}{365} \quad (2)$$

where, time-to-expiry T uses minutes from “now” to settlement, risk-free rate R uses the relevant NSE MIBOR tenor, the forward F comes from the same-expiry NIFTY futures.

We compute an India-VIX-style index using the variance-swap replication weights but along the lines of work in [6]. This involves selecting all OTM options each day with “near term” ($T_1 < T$) and “next term” ($T_2 > T$) expiry and calculating the corresponding weights,

$$\begin{aligned} \sigma_1^2 &= \frac{2}{T_1} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT_1} Q(K_i, T_1) - \frac{1}{T_1} \left(\frac{F}{K_0} - 1 \right)^2, \\ \sigma_2^2 &= \frac{2}{T_2} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT_2} Q(K_i, T_2) - \frac{1}{T_2} \left(\frac{F}{K_0} - 1 \right)^2. \end{aligned} \quad (3)$$

where, $\Delta K_i = K_{i+1} - K_i$, $\sum_i$ runs over the OTM options for each day. For example, say at a specific day t we have 10 OTM options, then

$$\sum_i \rightarrow \{K_{-10}, K_{-9}, \dots, K_0, K_1, K_2, \dots, K_{10}\} \quad \text{with} \\ K_{-10} < K_{-9} < K_0 < K_1 < \dots < K_{10} \quad (4)$$

Here, K_0 denotes the at-the-money (ATM) strike price on day t . It is selected as the strike closest to, but not exceeding, the underlying value at time t . $K_1, K_2, \dots$ are strike prices of Call option(CE) OTM's while $K_{-1}, K_{-2}, \dots$ denotes Put option(PE) OTM's. Combining all these VIX is computed as in eqn.(2)

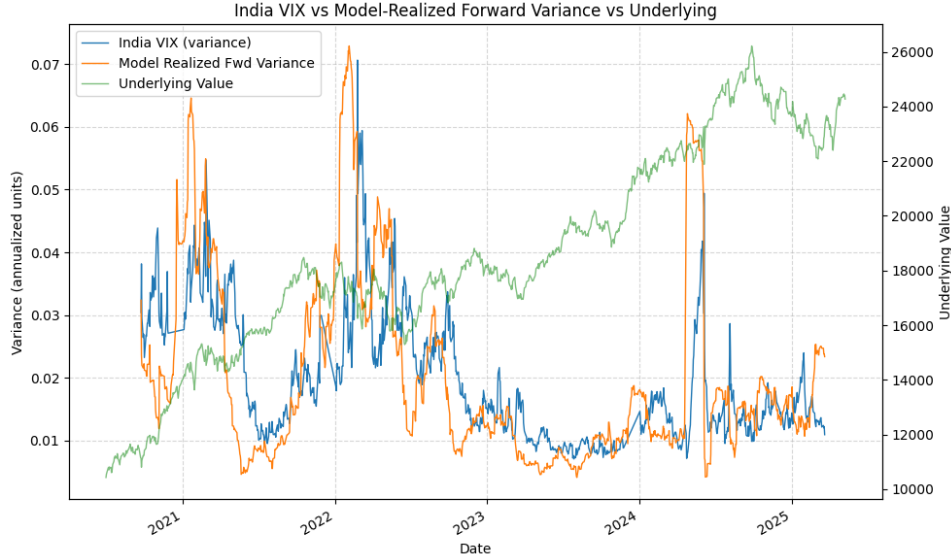


Figure 1: India VIX vs annualized variance of NIFTY 50

We plot the computed India VIX(orange) vs realized variance(blue, computed via (7)) above. The plot shows that realized variance and VIX almost run with each other. We also plot the underlying stock price(green) converted to a scale to match with VIX.

3 Regression problem setup

Our goal is to forecast future realized variance of returns. The realized return is defined as,

$$r_{t,l} = \frac{p_t - p_{t-l}}{p_{t-l}}, \quad l = 1 \quad (5)$$

where, p_t denotes the price of the underlying return at time t , l is the lookback horizon used to compute returns. In our case we use daily returns, i.e $l = 1$. One can use different values of $l = 2, 3, 4, \dots$. In our case we use log return

$$r_{t,l} = \ln\left(\frac{p_t}{p_{t-l}}\right) = \ln(p_t) - \ln(p_{t-l}) \quad (6)$$

in the regression problem. Now, the realized variance between time t and $t + T$ is defined as,

$$Var_t^T = \frac{1}{T} \sum_{i=1}^T (r_{t+i} - \bar{r})^2 \quad (7)$$

where, r_{t+i} is the daily return of the underlier, $\bar{r}$ is the mean return between time $t + 1$ and $t + T$ which is close to zero.

Now let us set up the regression problem. We fix two target variables calling them regression I and regression II. The regression equations are,

$$\text{Regression I} \quad Var_t^T = f(\{F_t(S, K, \tau)\}) + \epsilon_t \quad (8)$$

$$\text{Regression II} \quad Var_t^T - VIX_t^2 = VRP_t = f(\{F_t(S, K, \tau)\}) + \epsilon_t \quad (9)$$

where $F_t(S, K, \tau)$ denotes set of features at time t , S , K being the spot price and strike price at t and τ is the time to expiry. $F_t(S, K, \tau)$ is explicitly discussed in the subsequent section.

3.1 Feature Generation

Following the option-based approach of [6] and related work, we construct the following set of quantities.

$$\begin{aligned}
\text{ATM price} &= \frac{1}{2}(CE_{K_0} + PE_{K_0}) \\
\text{Skew1 price} &= (CE_{K_1} - PE_{K_{-1}}) \\
\text{Butterfly1 price} &= CE_{K_1} + PE_{K_{-1}} - 2\text{ATM price} \\
\text{ATM IV} &= \frac{1}{2}\left(IV(CE_{K_0}) + IV(PE_{K_0})\right) \\
\text{Skew1 IV} &= IV(CE_{K_1}) - IV(PE_{K_{-1}}) \\
\text{Butterfly1 IV} &= IV(CE_{K_1}) + IV(PE_{K_{-1}}) - 2\text{ATM IV} \\
\text{ATM TS} &= \frac{1}{2}(TS(CE_{K_0}) + TS(PE_{K_0})) \\
\text{Skew1 TS} &= TS(CE_{K_1}) - TS(PE_{K_{-1}}) \\
\text{Butterfly1 TS} &= TS(CE_{K_1}) + TS(PE_{K_{-1}}) - 2\text{ATM TS}
\end{aligned} \tag{10}$$

where,

$$\begin{aligned}
CE_{K_0} &= \text{price of the call option ATM,} \\
PE_{K_0} &= \text{price of the put option ATM,} \\
CE_{K_1} &= \text{price of the call option first OTM,} \\
PE_{K_{-1}} &= \text{price of the put option first OTM,} \\
IV(CE_{K_0}) &= \text{implied volatility(IV) of ATM CE,} \\
IV(PE_{K_0}) &= \text{IV of ATM PE,} \\
IV(CE_{K_1}) &= \text{IV of first OTM CE,} \\
IV(PE_{K_{-1}}) &= \text{IV of first OTM PE,} \\
TS &= 100(IV(T_1) - IV(T_2)), \\
TS(CE_{K_0}) &= \text{Term structure(TS) of ATM CE etc}
\end{aligned} \tag{11}$$

The price-based feature **ATM price** captures the market’s overall expectation of future volatility. The feature **Skew1 price** quantifies the asymmetry between slightly OTM call and put options, where a positive skew generally signals an expectation of upward market movement. Finally, **Butterfly1 price** characterizes the curvature, or “U-shape,” of the option price profile.

Now, moving to the IV-based features, **ATM IV** is closely related to the square root of the expected future variance and is therefore directly predictive of RV. In contrast, **Skew1 IV** captures the difference in IV between slightly OTM calls and puts. Finally, **Butterfly1 IV** quantifies the “fat-tailedness” or curvature of the IV distribution.

Now, **ATM TS** says the slope of IV over time at ATM strike, where **Skew1 TS** show skew changes across maturities and **Butterfly1 TS** measures curvature of the term structure across strikes.

We use the z-score of these quantities as features in our regression models (8) and (9). However depending on the model the p-value of these features changes. Features with higher p-values have been discarded. These issues are discussed in detail in the following sections.

In our dataset, however, we find that the OTM option prices exhibit a high degree of linear dependence across adjacent strikes, leading to multicollinearity problems in the regression design matrix. To quantify this effect, we compute the Variance Inflation Factor (VIF) for first five OTM “Close” series on each day given in 1. All the features display VIF values well above 10, confirming severe multicollinearity among neighboring strikes. One way to circumvent this problem is to do principal component analysis(PCA). However we use the above discussed features in (10) to avoid the problem of multicollinearity as can be seen from 2, where VIF’s are reduced as opposed to 1.

Feature	VIF
CE_{K_0}	27.698490
CE_{K_1}	41.520776
CE_{K_2}	26.723511
CE_{K_3}	27.477633
CE_{K_4}	21.104153
CE_{K_5}	24.663396
PE_{K_0}	22.271485
$PE_{K_{-1}}$	18.791581
$PE_{K_{-2}}$	19.020102
$PE_{K_{-3}}$	12.617817
$PE_{K_{-4}}$	13.921355
$PE_{K_{-5}}$	16.929175

Table 1: VIF for raw option prices.

Feature	VIF
Skew1_price	15.153547
ATM_price	13.460925
Skew1_IV	13.059614
Butterfly1_IV	9.002695
Butterfly1_price	8.550346
ATM_IV	7.502156
annual_bwd_var	3.870393
Butterfly1_TS	1.700843
ATM_TS	1.389842
Skew1_TS	1.154499

Table 2: VIF for constructed option-based features.

These constructed features are used as inputs to all three regression models — linear regression, ridge regression, and feed-forward neural network (FNN) — ensuring consistent and numerically stable learning across methods.

3.2 Model adequacy and residual analysis

In this section we briefly examine the following aspects of these features in the context of fitting a linear regression model.

1. Non-linearity of the response-predictor relationships
2. Correlation of error terms
3. Non-constant variance of error terms
4. Outliers
5. Collinearity

1. Non-linearity of the data:

We check if there is non linearity present in the data by fitting residuals $y - \hat{y}$ vs fitted $\hat{y}$ values of the response variable y . The blue line is a smooth fit to the residuals. The U-shape of this fit indicates non-linear associations of the predictor and response. In such cases it is useful to make a non-linear transformation of the predictors variable, such as $\log X$, $\sqrt{X}$, X^2 etc. After taking log of the target we find that the trend is almost removed.

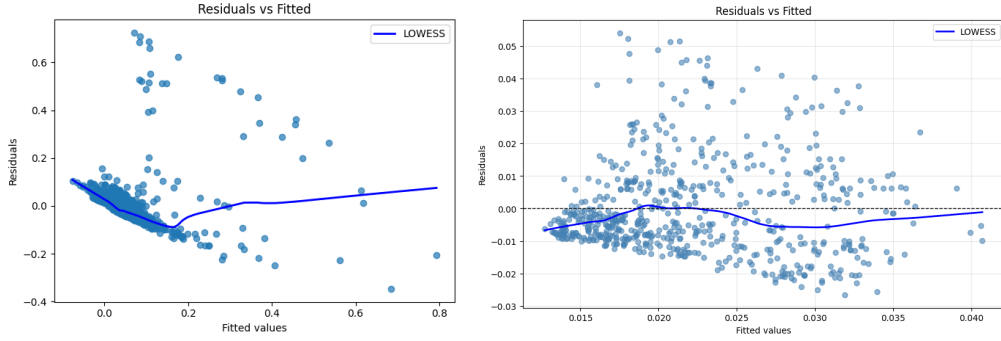


Figure 2: Left is Residual vs fitted values for $y = Var_t$. The right plot is for $y = \log Var_t$

2. Auto correlation of error terms:

In a time series data if there exists a correlation among the error terms there will be a trend present in the plot of residuals over time. The correlation between r_t and r_{t-l} is called lag- l auto-correlation function of r_t and is defined by

$$\rho_l = \frac{Cov(r_t, r_{t-l})}{Var(r_t)} \quad (12)$$

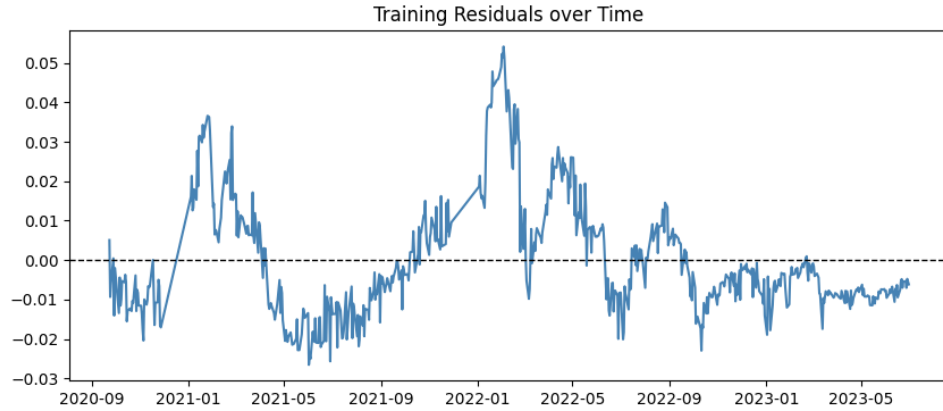


Figure 3: Training residuals of the linear regression model over time. The residuals exhibit prolonged runs of positive and negative errors, suggesting temporal dependence and heteroskedasticity.

To see if the regression models capture time patterns in the data, we check whether the model residuals show any serial correlation. Ideally, the residuals should behave as white noise exhibiting zero mean, constant variance, and no serial correlation. Figure 3 plots the training residuals of the linear regression model over time. While the residuals fluctuate around zero, they display prolonged periods of positive and negative values.

To further check the correlation of errors in our time series data we compute autocorrelation function (ACF) of the training residuals and plot in 4.

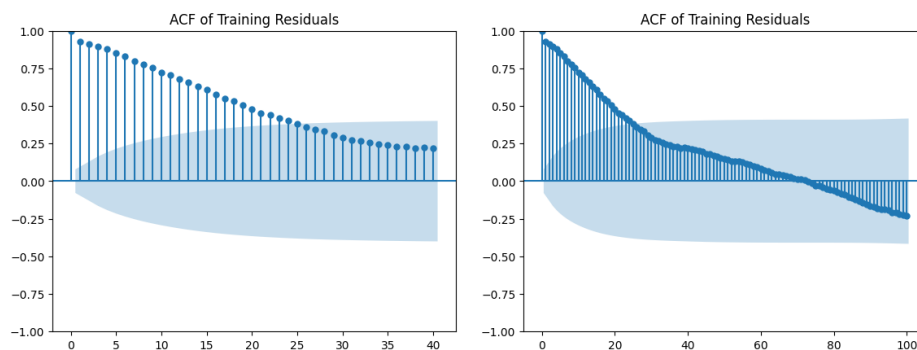


Figure 4: Autocorrelation function (ACF) of training residuals for the linear regression model. The slowly decaying positive correlation indicates serial dependence.

The ACF exhibits a slow exponential decay, remaining significantly positive up to at least 40 lags. Such a pattern indicates strong positive autocorrelation in the residuals, implying that the model errors are serially dependent.

3. Non-constant variance of error terms:

To examine whether the model errors exhibit constant variance, we analyze the residuals as a function of fitted values. Figure 5 plots the residuals of the linear regression model against the corresponding fitted values. Each point represents an observation in the training data, with the residuals defined as $\varepsilon_i = y_i - \hat{y}_i$.

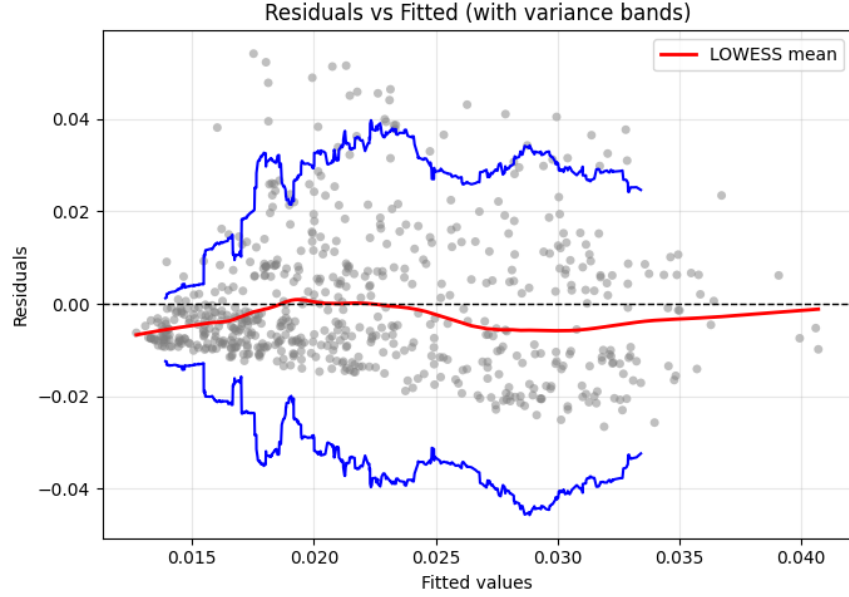


Figure 5: Residuals versus fitted values with LOWESS mean (red) and smoothed $\pm 2\sigma$ variance bands (blue). The widening spread of residuals indicates heteroskedasticity, consistent with volatility clustering in financial data.

The red line shows the locally weighted scatterplot smoothing (LOWESS) of the residuals, capturing the mean trend, while the blue curves represent the smoothed $\pm 2\sigma$ variance bands obtained from a rolling standard deviation of residuals. Ideally, for homoskedastic errors, the residuals should be randomly scattered around zero with a roughly constant spread across all fitted values.

However, the plot shows a funnel-shaped pattern: the spread of residuals increases with higher fitted values. This indicates the presence of heteroskedasticity, where the variance of errors depends on the magnitude of the predicted variable.

In the context of realized variance and India VIX prediction, such behavior is expected. Periods of higher volatility tend to display larger

uncertainty in both observed and predicted variance, leading to greater residual dispersion.

4. **Outliers:** The points for which y_i is far from its predicted value are called outliers. We compute studentized residuals to identify outliers. We remove all those points for which the absolute value of studentized residual is greater than 2. To identify potential outliers and influential observations, we compute the externally studentized residuals for the fitted model. For each observation i , the studentized residual is defined as

$$t_i = \frac{\varepsilon_i}{s_i \sqrt{1 - h_{ii}}},$$

where ε_i is the raw residual, h_{ii} is the leverage of the i -th observation, and s_i is the standard deviation of the residual computed with the i -th observation omitted. Under the null hypothesis $H_0 : \varepsilon_i = 0$, the statistic t_i follows a t -distribution with $(n - p - 1)$ degrees of freedom, where n is the sample size and p the number of model parameters. Figure 6 plots the studentized residuals against the fitted values.

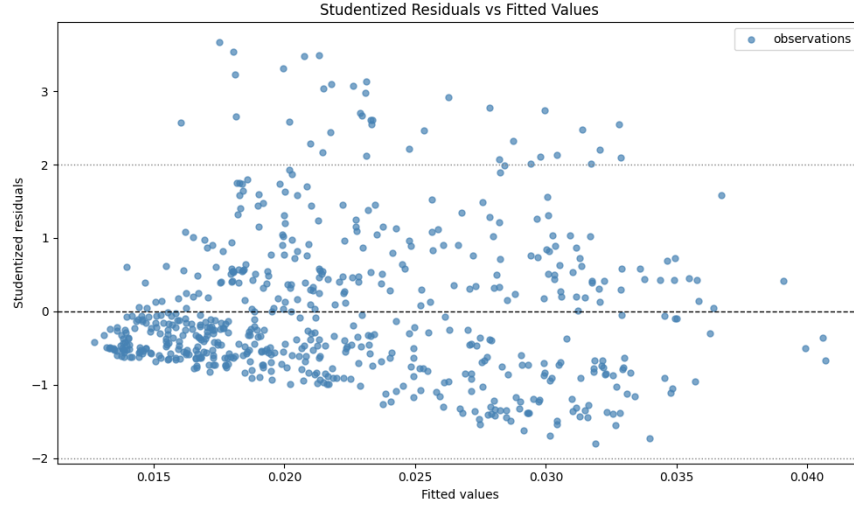
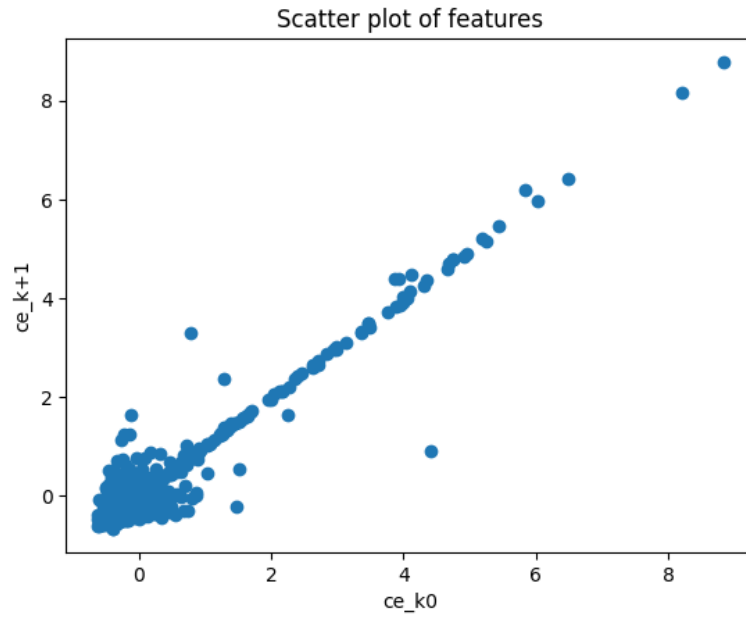
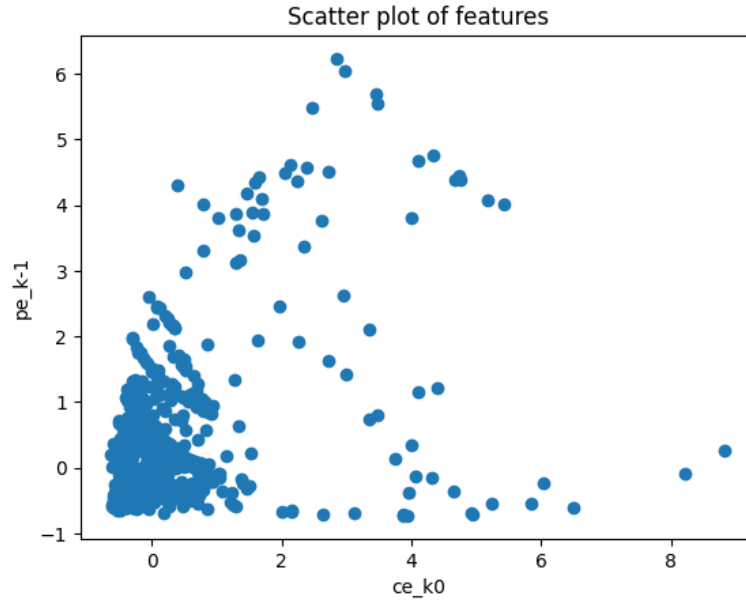


Figure 6: Studentized residuals versus fitted values. The dashed black line marks zero, and the dotted gray lines denote the ± 2 thresholds. Observations with $|t_i| > 2$ are identified as outliers and removed prior to re-estimation.

Most residuals lie within the ± 2 bounds (indicated by gray dotted lines), suggesting that the model captures the majority of the data well. However, a small number of observations exhibit $|t_i| > 2$, indicating statistically significant deviations from the fitted model. We treat these points as potential outliers and remove them before performing the final model fitting and analysis. The process is applied consistently across all three models.

5. **Multi-collinearity:** We plot scatter diagrams of some the raw option price features. We observe that call option OTM's have linear relationship among them. Similarly put OTM's are also linear with each other. However call and put option OTM's do not have an obvious relationship. We present some of the scatter plots below.



We examine the pairwise correlations among the features in (10) using the heatmap in Figure 7. The plot reveals strong correlations between ATM IV

and ATM price, between Skew1 IV and Skew1 price, and between Butterfly1 IV and Butterfly1 price. To avoid multicollinearity, we exclude these highly correlated feature pairs during model estimation, taking their statistical significance (p-values) into account.

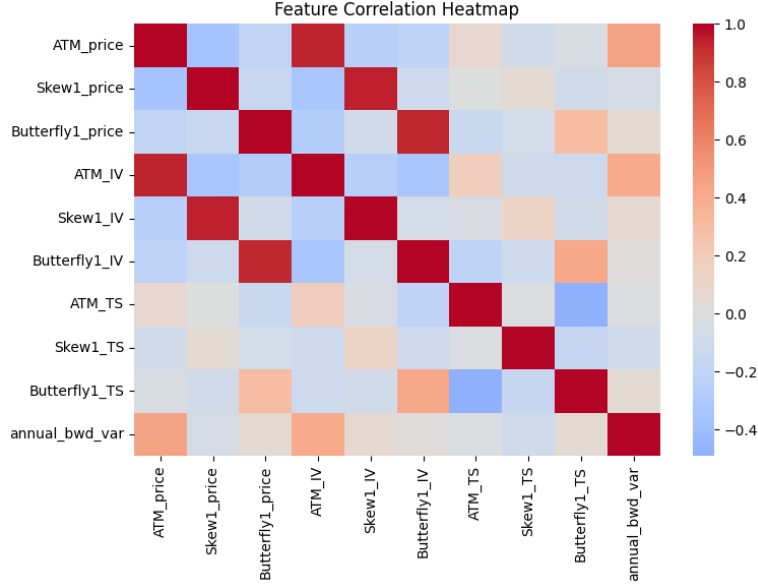


Figure 7: heatmap

4 Machine learning models

4.1 Ordinary least square(OLS)

In this section we briefly review OLS. Given a response variable $y_i, i = 1, 2, \dots, n$ and a set of predictors $X \in \mathbb{R}^{n \times p}$, the OLS model assumes a linear relationship of the form

$$y = X\beta + \varepsilon, \quad (13)$$

where $\beta \in \mathbb{R}^p$ is the vector of regression coefficients and ε is an error term with mean zero. The OLS estimator $\hat{\beta}$ is obtained by minimizing the sum of

squared residuals,

$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad (14)$$

provided that $X^\top X$ is invertible. This solution chooses the coefficients that best fit the data in a least-squares sense.

In our setting, the predictors X consist of the features in (10) and the targets are either the log realized variance $\ln Var_t$ or the variance risk premium VRP_t . The OLS model provides a simple setting to conduct the residual analysis in section 3.2 that facilitates in checking assumptions of linear regression. However, when predictors are highly correlated OLS can become unstable, motivating the use of regularized extensions such as Ridge and Elastic Net.

4.2 Ridge

Ridge Regression is a regularized version of the ordinary least squares (OLS) regression that addresses the problem of multicollinearity among predictors by imposing an ℓ_2 penalty on the magnitude of regression coefficients. The Ridge estimator is obtained by minimizing:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \alpha \sum_{j=1}^p \beta_j^2 \quad (15)$$

where $\alpha > 0$ is the regularization parameter controlling the strength of shrinkage. α is tuned such that the mean squared error of the test data is minimum. The ℓ_2 penalty term $\sum_j \beta_j^2$ discourages large coefficient values by shrinking them toward zero. Consequently, all features remain in the model, but their influence is reduced. This makes Ridge regression particularly effective in scenarios where the predictors are highly correlated or when $p > n$ (i.e., more predictors than observations).

4.3 Elastic Net Regression

The Elastic Net [15] is a regularized linear regression model that combines the penalties of both Lasso (ℓ_1) and Ridge (ℓ_2) regression. It is particularly effective when the predictors are highly correlated or when the number of predictors exceeds the number of observations. The Elastic Net estimator minimizes the following function:

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \alpha \sum_{j=1}^p \beta_j^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (16)$$

When $\alpha = 0$, the model reduces to the Lasso regression, and when $\lambda = 0$, it becomes the Ridge regression. The ℓ_2 penalty (with weight α) helps stabilize the estimates when predictors are highly correlated, while the ℓ_1 penalty (with weight λ) performs variable selection by shrinking some coefficients exactly to zero. As a result, the Elastic Net balances sparsity and stability, making it especially useful in settings with multicollinearity or when the number of predictors is large.

4.4 Feedforward Neural Network (FNN)

A feedforward neural network (FNN) is a nonlinear model that takes an input vector $(X_1, X_2, \dots, X_p)$ of features and builds a nonlinear function $f(X)$ to predict the response y .

$$f(X) = \beta_0 + \sum_{k=1}^K \beta_k g(w_{k0} + \sum_{j=1}^p w_{kj} X_j) \quad (17)$$

w_{kj} 's are called weights associated to the connection between j -th input node to k -th node of a hidden layer. The above is a single layer FNN. Information in an FNN flows only forward through the layers, with no feedback connections. The input variables form the input layer, and their weighted combinations are passed into a set of hidden units, where each hidden unit

applies a nonlinear activation function such as ReLU. These hidden-layer activations are then combined to produce the final prediction. The network is trained by minimizing a regularized loss:

$$\mathcal{L}(\beta, w) = \frac{1}{n} \sum_{i=1}^n \ell \left(y_i, \beta_0 + \sum_{k=1}^K \beta_k g \left(w_{k0} + \sum_{j=1}^p w_{kj} X_{ij} \right) \right) + \lambda \left(\sum_{k=1}^K \beta_k^2 + \sum_{k=1}^K \sum_{j=1}^p w_{kj}^2 \right). \quad (18)$$

where ℓ is either Mean Squared Error (MSE) or the Huber loss with threshold δ . Optimization is performed via stochastic gradient descent method. In our setup, the FNN has two hidden layers of sizes (32, 16) with ReLU activations and a linear output.

5 Results

OLS:

We list the coefficients of regression for regression I and II below in table 3 and 4 respectively. We keep only those features for which p-value is very small and t-statistic is significantly large. By doing so we find that for regression I and II different set of features are significant for the regression.

Variable	Coefficient	Std. Error	t-Statistic	p-value
const	0.0222	0.001	37.175	0.000
ATM_IV	0.0065	0.001	9.090	0.000
Skew1_IV	0.0020	0.001	3.163	0.002
Butterfly1_IV	0.0035	0.001	5.043	0.000
Skew1_TS	-0.0019	0.001	-3.019	0.003

Table 3: Regression coefficients and significance statistics for regression I.

Variable	coefficient	Std err	t-Statistic	p-value
const	0.1273	0.017	7.408	0.000
ATM_price	0.4039	0.060	6.787	0.000
Skew1_price	0.2730	0.063	4.300	0.000
ATM_IV	-0.3172	0.059	-5.411	0.000
Skew1_IV	-0.2472	0.062	-3.976	0.000
Butterfly1_TS	-0.0616	0.018	-3.468	0.001

Table 4: Regression coefficients and significance statistics for regression II.

We note down the in sample and out sample performances for linear regression models. We find that regression II perform poorly compared to regression I. This is expected because in regression II we have VIX, which is noisy and difficult to model.

$y = \ln Var_t$	R^2	RMSE	Correlation($y, \hat{y}$)
Train	0.146	0.01487	0.382
Test	0.063	0.01237	0.325

$y = VRP_t$	R^2	RMSE	Correlation($y, \hat{y}$)
Train	0.063	0.0128	0.25
Test	-0.154	0.00608	0.051

Ridge regression:

The Ridge regression results show a clear improvement over the linear model for both target variables. For the VRP forecast $y = VRP_t$, introducing the regularization parameter $\alpha = 0.001$ increases the out-of-sample R^2 from negative values (in the linear model) to a positive 0.012, accompanied by a stronger correlation of 0.225 between the predicted and realized values. This indicates that Ridge successfully reduces overfitting by stabilizing the coefficient estimates through the bias-variance tradeoff.

$y = \ln Var_t$	α	R^2	RMSE	Correlation($y, \hat{y}$)
Train	0.001	0.138	0.01494	0.378
Test	0.001	0.060	0.01240	0.346

$y = VRP_t$	α	R^2	RMSE	Correlation
Train	0.001	0.028	0.01307	0.182
Test	0.001	0.012	0.0056	0.225

Elastic net:

For the Elastic Net model, we observe that introducing a small amount of mixed ℓ_1 - ℓ_2 regularization ($\alpha = 0.01$, $L_1 = 0.001$) yields meaningful improvements in predicting $y = \ln Var_t$. The in-sample performance shows an R^2 of 0.146 and a correlation of 0.382, while the out-of-sample performance remains stable with an R^2 of 0.060 and a correlation of 0.327. These results are almost identical to those obtained from the Ridge model, suggesting that the ℓ_1 penalty is small enough that the model behaves very similarly to pure ℓ_2 regularization. The fact that the out-of-sample R^2 remains positive indicates that Elastic Net successfully captures some predictive features without overfitting.

In contrast, the Elastic Net model performs noticeably worse for the variance risk premium target $y = VRP_t$. Although the in-sample performance shows moderate predictive power ($R^2 = 0.069$, correlation 0.265), the out-of-sample metrics deteriorate significantly, with a negative R^2 of -0.105 and a very weak correlation of 0.081. This indicates that the model is unable to generalize to unseen data for VRP prediction. The poor performance suggests that the VRP series is either noisier or less directly linked to the option-based features used here. The poor performance is also a result of less available data, we have only 885 data point in training set and 112 in the test set. As a result, regularization does not help much, and the model overfits the training data despite small penalties. Overall, Elastic Net provides reasonable predictive performance for $\ln Var_t$, but its ability to forecast the variance risk premium is limited.

$y = \ln Var_t$	α	L_1	R^2	RMSE	Correlation($y, \hat{y}$)
Train	0.01	0.001	0.146	0.01488	0.382
Test	0.01	0.001	0.060	0.01239	0.327

$y = \ln VRP_t$	α	L_1	R^2	RMSE	Correlation($y, \hat{y}$)
Train	0.01	0.001	0.069	0.50951	0.265
Test	0.01	0.001	-0.105	0.35855	0.081

FNN:

We present the results of the FNN model below for both the regressions. FNN performs poorly compared to simpler models such as OLS and Ridge primarily due to the limited sample size of our dataset (total 997 observations). Neural networks are high-capacity models that require large amounts of data to generalize well. In small-sample settings with noisy financial features, they tend to overfit the training data and fail to capture stable relationships. In contrast, OLS and Ridge impose stronger structural constraints, making them more robust and effective for small, noisy datasets.

$y = \ln \text{Var}_t$	R^2	RMSE	Corr($y, \hat{y}$)
Train	-0.038	0.016399	-0.089
Test	-0.261	0.014357	-0.017

$y = \ln VRP_t$	R^2	RMSE	Corr($y, \hat{y}$)
Train	0.033	0.519156	0.193
Test	-0.309	0.390299	0.225

6 Conclusion

In this paper, we study how option-implied information can be used to forecast volatility for the NIFTY index and to understand the relation between realized variance and the India VIX. We first review the India VIX construction and implement an India-VIX-style index using NIFTY option data. We then define two prediction targets: future realized variance and the variance risk premium, given by the difference between realized variance and India VIX squared.

To address strong multicollinearity in raw option prices, we construct a small set of features based on ATM and first OTM options: level (**ATM**), skew (**Skew1**), and curvature (**Butterfly1**) in prices, implied volatilities, and term structure. We show that these engineered features have reduced VIF values and provide a more suitable input set for regression. We also carry out residual diagnostics (non-linearity, autocorrelation, heteroskedasticity and outliers) and apply basic cleaning steps such as log-transforming the target and removing extreme observations.

Using these features, we compare several supervised learning models: ordinary least squares, Ridge regression, Elastic Net, and a feedforward neural network. For the realized variance target $y = \ln Var_t$, linear and Ridge-type models achieve modest but positive out-of-sample R^2 and meaningful correlations. For the variance risk premium $y = VRP_t$, the forecasting problem is much harder: Ridge offers a small improvement over plain OLS, but overall out-of-sample performance remains weak. The neural network does not outperform the simpler models, mainly due to the limited sample size and the noisy nature of financial volatility.

Overall, our results suggest that for India VIX and NIFTY volatility, simple regularized linear models with carefully designed option-implied features are more reliable than more complex machine learning architectures at the current data scale. Future work could extend this framework by using longer samples, intraday or high-frequency data, richer term-structure features, and dynamic models that better capture the time-series dependence in volatility and residuals.

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